Software Requirements Specification

Online Exam Portal

Prepared according to IEEE Std 830-1998 / ISO/IEC/IEEE 29148 structure

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Revision History

Version	Date	Author	Change Summary
1.0	03-10-2026	Chhavi (P1)	Initial draft of complete SRS
1.1	03-10-2026	Chhavi (P1)	Technology stack fixed as MERN (sections 1.3, 2.1, 2.5, 3.1.3, 3.6, Appendix B); NoSQL injection added to SEC-OE-008
1.2	07-10-2026	Chhavi (P1)	Result-release rule added (2.6); RTM completed with test case IDs (STP) and components (SAD); open issues closed

Approvals

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional, non-functional and security requirements of the Online Exam Portal, a web-based system for conducting multiple-choice question (MCQ) examinations with automatic evaluation. It is the baseline for the Software Test Plan (STP) and the Software Architecture & Design Specification (SAD) of this project, and is intended for developers, testers, instructors reviewing the project and the project team.

1.2 Scope

The Online Exam Portal lets Instructors build a question bank and schedule timed MCQ exams, lets Students take assigned exams in a web browser, evaluates attempts automatically, and lets Students and Instructors view results. Administrators manage users and review audit logs.

In scope: account management, question bank, exam creation and scheduling, timed exam attempt with auto-save, auto-submission, automatic evaluation, results and basic analytics, notifications, focus-loss logging, audit logging.

Out of scope (this release): descriptive or coding questions, video/webcam proctoring, mobile native apps, payment, integration with external learning management systems.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
STP	Software Test Plan
SAD	Software Architecture & Design Specification
RTM	Requirements Traceability Matrix
MCQ	Multiple-choice question with exactly one correct option
FR / NFR / SEC	Functional / Non-functional / Security requirement
RBAC	Role-Based Access Control
TLS	Transport Layer Security
OWASP	Open Web Application Security Project
WCAG	Web Content Accessibility Guidelines
Attempt	One Student's single run of an exam, from start to submission
Exam window	Period between the exam's start and end date-time during which an attempt may begin
MERN	MongoDB, Express.js, React, Node.js – the JavaScript technology stack used for the system
Question bank	Repository of MCQs owned by Instructors, from which exams are assembled

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.

- ISO/IEC/IEEE 29148:2018, Systems and software engineering – Life cycle processes – Requirements engineering.
- OWASP Top 10 – Web Application Security Risks; OWASP ASVS.
- W3C, Web Content Accessibility Guidelines (WCAG) 2.1.
- Project documents: Software Test Plan (STP) and Software Architecture & Design Specification (SAD) of the Online Exam Portal.

1.5 Document Overview

Section 2 gives an overall description of the product. Section 3 holds the specific requirements: external interfaces, functional requirements, the use-case model, non-functional requirements and the security section. Section 4 contains the initial traceability matrix. Section 5 contains appendices. Requirement IDs follow the convention in Appendix A and are frozen once agreed by the team.

2. Overall Description

2.1 Product Perspective

The Online Exam Portal is a new, self-contained web application accessed through a browser. It follows the MERN stack: a React web front end, a Node.js/Express back-end application server exposing a REST API, and a MongoDB database. It interacts with one external system: an email service (SMTP) used for account verification, password reset and notifications.

2.2 Product Functions

- Account registration, login, logout and password reset.
- User and role administration.
- Question bank and exam management by Instructors.
- Exam publication and assignment to Students, with notification.
- Timed exam attempt with auto-save, resume, manual submission and automatic submission.
- Automatic evaluation, result display, analytics and CSV export.
- Focus-loss logging and audit logging.

2.3 User Classes and Characteristics

User class	Description	Technical expertise / frequency
Student	Registers, takes assigned exams, views own results.	Basic web skills; uses system around exam dates; up to 500 simultaneously.
Instructor	Creates questions and exams, assigns exams, views results.	Moderate; regular use.
Administrator	Manages users and roles, reviews audit logs.	Moderate to high; occasional use.
Email Service	External system that delivers emails sent by the portal.	Automated.

2.4 Operating Environment

- Client: latest two major versions of Google Chrome, Mozilla Firefox and Microsoft Edge on Windows, macOS, Linux or Android.
- Server: Linux server or container with a database server and HTTPS support.
- Network: Internet or institutional intranet connection of at least 1 Mbps per Student.

2.5 Design and Implementation Constraints

- The system shall be a web application built on the MERN stack: MongoDB (database), Express.js (server framework), React (front end) and Node.js (runtime). Component and library choices are detailed in the SAD.
- The system shall follow OWASP Top 10 secure-coding practices and shall use only open-source libraries compatible with academic use.
- The system shall be completed as an academic mini-project within the schedule ending 07-10-2026.

2.6 Assumptions and Dependencies

- Students have a working device, browser and Internet connection during exams.
- An SMTP email service is available and reachable by the server.
- Instructors supply exam questions and the list of Students for each exam.
- Only MCQs with a single correct option are supported in this release.
- Result release (clarifies FR-OE-023): a Student sees the score and pass/fail status immediately after submission; the question-wise review with correct answers is shown only after the exam end date-time and only if the Instructor has enabled it.

2.7 Apportioning of Requirements

Requirements marked Could are candidates for deferral if the schedule is at risk. Webcam-based proctoring, descriptive-answer evaluation and mobile apps are deferred to future releases.

3. Specific Requirements

Requirement-writing convention. Each requirement uses the word “shall”, states one verifiable behaviour, and gives a measurable value where applicable. Priority uses MoSCoW: Must (essential), Should (important, deferrable), Could (desirable).

3.1 External Interface Requirements

3.1.1 User Interfaces

- The system shall provide a responsive web interface with role-specific dashboards for Student, Instructor and Administrator.
- The exam page shall show the question, its options, a question navigator, the remaining time and a Submit button at all times.

3.1.2 Hardware Interfaces

None. The system runs on standard server hardware and client devices with a keyboard or touch screen.

3.1.3 Software Interfaces

- Email Service: the system shall send email through SMTP for verification, password reset and notifications.
- Database: the system shall store all persistent data in MongoDB, accessed only by the back-end server.

3.1.4 Communication Interfaces

- Browser to server communication shall use HTTPS and a REST/JSON API (see SEC-OE-005).

3.2 Functional Requirements

3.2.1 Account & Access

ID	Requirement	Priority	Acceptance criterion
FR-OE-001	The system shall allow a visitor to register a Student account by providing full name, a unique email address and a password.	Must	Registration with a unique email creates an account; a duplicate email is rejected with an error message.
FR-OE-002	The system shall verify a new account by sending a verification link to the registered email address; the link shall remain valid for 24 hours.	Must	Login is blocked until the link is clicked; a link older than 24 hours is rejected.
FR-OE-003	The system shall authenticate users by email and password and shall grant access only to the features permitted for the user's role (Student, Instructor or Administrator).	Must	Valid credentials open the role-specific dashboard; invalid credentials show a generic error and no access.
FR-OE-004	The system shall allow a logged-in user to log out, which shall invalidate the current session immediately.	Must	After logout, the back button or a saved URL cannot access protected pages.
FR-OE-005	The system shall allow a user to reset a forgotten password through a one-time link sent by email, valid for 30 minutes.	Must	A used or expired link is rejected; the new password replaces the old one.
FR-OE-006	The system shall allow an Administrator to create, edit, activate, deactivate and assign roles to user accounts.	Must	A deactivated user cannot log in; a role change applies from the user's next login.

3.2.2 Exam Setup

ID	Requirement	Priority	Acceptance criterion
FR-OE-007	The system shall allow an Instructor to create, edit and delete multiple-choice questions, each with a question text, 2 to 6 options, exactly one correct option and a mark value.	Must	A question with zero or more than one correct option cannot be saved.
FR-OE-008	The system shall allow an Instructor to create an exam by specifying title, duration in minutes, start date-time, end date-time, pass mark and the set of questions selected from the question bank.	Must	An exam cannot be saved if the end time is not after the start time or if it has no questions.

ID	Requirement	Priority	Acceptance criterion
FR-OE-009	The system shall allow an Instructor to enable or disable shuffling of question order and option order for an exam.	Should	With shuffling on, two students receive different question orders in at least 90% of sampled pairs.
FR-OE-010	The system shall allow an Instructor to configure optional negative marking (a deduction value per wrong answer) for an exam.	Could	A wrong answer deducts the configured value; an unanswered question deducts nothing.
FR-OE-011	The system shall allow an Instructor to publish an exam and assign it to selected Students, after which the exam appears on those Students' dashboards.	Must	Only assigned Students can see the exam; an unpublished exam is visible to nobody but its Instructor.
FR-OE-012	The system shall prevent editing of questions, marks and duration of an exam after the first Student has started an attempt.	Must	Edit attempts on a started exam are rejected with an explanatory message.

3.2.3 Exam Taking

ID	Requirement	Priority	Acceptance criterion
FR-OE-013	The system shall list to each Student the exams assigned to them with title, duration, time window and status (Upcoming, Open, Attempted, Missed).	Must	Status is correct against the current server time.
FR-OE-014	The system shall allow a Student to start an assigned exam only between its start and end date-times and only once.	Must	A second start attempt, or a start outside the window, is rejected.
FR-OE-015	The system shall present questions to the Student, allow selecting or changing one option per question, navigating freely between questions and flagging a question for review.	Must	The selected option and flag persist when navigating between questions.
FR-OE-016	The system shall display a countdown timer with the remaining time, calculated from the server clock, throughout the attempt.	Must	Displayed time differs from server time by at most 2 seconds.
FR-OE-017	The system shall automatically save the Student's answers to the server every 30 seconds and whenever an answer is changed.	Must	After a browser crash, all answers saved before the crash are restored.
FR-OE-018	The system shall allow a Student to resume an interrupted attempt, with the timer continuing from the original end time of the attempt.	Should	Resumed attempt shows previously saved answers and no extra time.
FR-OE-019	The system shall allow a Student to submit the exam manually after a confirmation prompt that shows the number of unanswered questions.	Must	Submission is final; further changes are rejected.
FR-OE-020	The system shall automatically submit the Student's saved answers when the timer reaches zero.	Must	At timer expiry the attempt is closed and submitted within 5 seconds.
FR-OE-021	The system shall record every event in which the exam window loses focus or the browser tab is switched, with a timestamp, and make the count and times visible to the Instructor.	Should	Each tab switch during an attempt appears in the Instructor's attempt report.

3.2.4 Evaluation & Results

ID	Requirement	Priority	Acceptance criterion
FR-OE-022	The system shall automatically evaluate each submitted attempt by comparing answers with the stored correct options and computing the total score using the question marks and any negative marking.	Must	Computed score matches a manual calculation for a test set of 20 attempts.
FR-OE-023	The system shall allow a Student to view their score, pass/fail status and, if enabled by the Instructor, a question-wise review with correct answers.	Must	A Student can view only their own results.
FR-OE-024	The system shall allow an Instructor to view, for each exam, the list of attempts with scores and summary statistics (average, highest, lowest score and pass percentage).	Must	Statistics match values computed independently from the attempt data.
FR-OE-025	The system shall allow an Instructor to export the results of an exam as a CSV file.	Should	The CSV contains one row per attempt and opens correctly in a spreadsheet application.
FR-OE-026	The system shall send an in-app notification and an email to Students when an exam is assigned to them and when their result is available.	Should	Notification is delivered within 2 minutes of the triggering event.
FR-OE-027	The system shall allow an Administrator to view an audit log of login events, exam creation, publication, submissions and user-management actions, filterable by date and user.	Should	Filtering by user and date range returns only matching entries.

3.3 Use-Case Model

Figure 1 shows the use-case diagram. Actors are Student, Instructor and Administrator; the Email Service is a secondary (external) actor.

Figure 1: UML Use-Case Diagram – Online Exam Portal

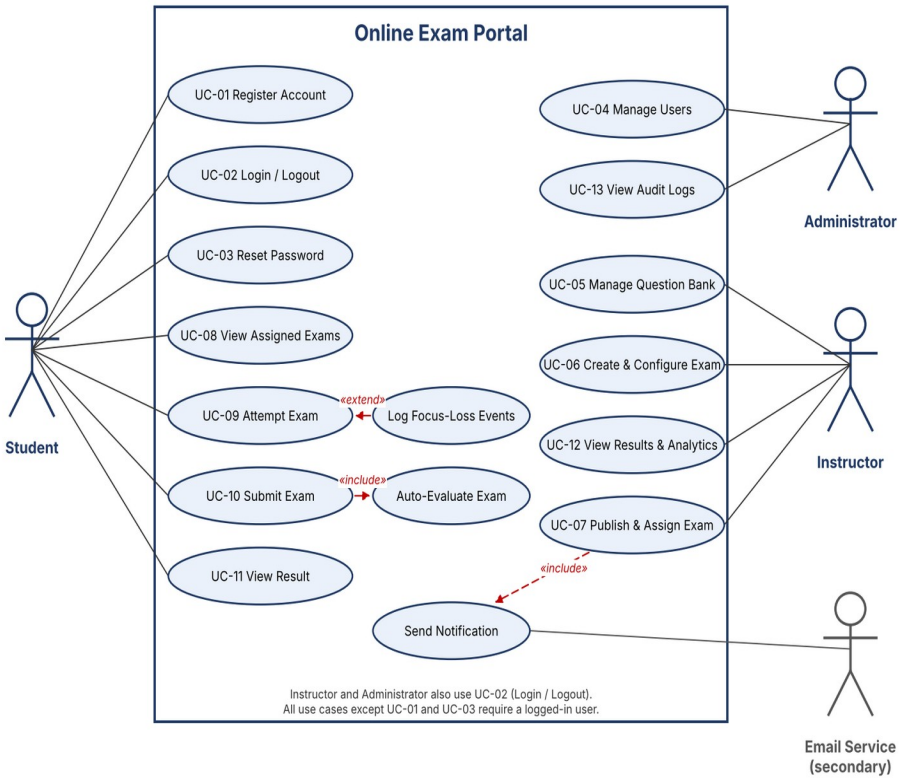


Figure 1: UML Use-Case Diagram – Online Exam Portal

Table 3.3 lists each use case with the functional requirements it realises.

ID	Use case	Actors	Requirements	Description
UC-01	Register Account	Visitor/Student	FR-OE-001, FR-OE-002	A visitor creates a Student account and verifies the email address.
UC-02	Login / Logout	Student, Instructor, Administrator	FR-OE-003, FR-OE-004	A user authenticates and ends the session. All use cases other than UC-01 and UC-03 require a logged-in user.
UC-03	Reset Password	All users, Email Service	FR-OE-005	A user sets a new password using an emailed one-time link.
UC-04	Manage Users	Administrator	FR-OE-006	The Administrator creates, edits, deactivates users and assigns roles.
UC-05	Manage Question Bank	Instructor	FR-OE-007	The Instructor adds, edits and deletes MCQs.
UC-06	Create & Configure Exam	Instructor	FR-OE-008, FR-OE-009, FR-OE-010, FR-OE-012	The Instructor defines an exam, its questions, time window and options.
UC-07	Publish & Assign Exam	Instructor, Email Service	FR-OE-011, FR-OE-026	The Instructor publishes the exam to selected Students, who are notified.
UC-08	View Assigned Exams	Student	FR-OE-013	The Student sees assigned exams and their status.

ID	Use case	Actors	Requirements	Description
UC-09	Attempt Exam	Student	FR-OE-014 to FR-OE-018, FR-OE-021	The Student answers questions under a running timer with auto-save and focus-loss logging.
UC-10	Submit Exam	Student	FR-OE-019, FR-OE-020, FR-OE-022	The Student submits manually or automatically at timer expiry; the system evaluates the attempt.
UC-11	View Result	Student	FR-OE-023	The Student views their own score and review.
UC-12	View Results & Analytics	Instructor	FR-OE-024, FR-OE-025, FR-OE-021	The Instructor views attempts, statistics and focus-loss reports and exports CSV.
UC-13	View Audit Logs	Administrator	FR-OE-027	The Administrator reviews security and activity logs.

Table 3.3: Use-case summary

3.4 Non-Functional Requirements

ID	Category	Requirement	Priority	Verification
NFR-OE-001	Performance	The system shall respond to 95% of page requests within 2 seconds under a load of 500 concurrent users.	Must	Load test
NFR-OE-002	Performance	The system shall complete an answer auto-save request in 1 second or less for 95% of requests at 500 concurrent users.	Must	Load test
NFR-OE-003	Performance	The system shall evaluate a submitted attempt and make the score available within 5 seconds of submission, for up to 500 simultaneous submissions.	Must	Load test
NFR-OE-004	Availability	The system shall be available at least 99.5% of the time during scheduled exam windows, measured per calendar month.	Must	Monitoring report
NFR-OE-005	Reliability	The system shall not lose more than 30 seconds of a Student's answers in the event of a client or server failure (recovery point objective of 30 seconds).	Must	Fault-injection test
NFR-OE-006	Reliability	The system shall take an automated backup of all data at least once every 24 hours and retain the last 7 backups.	Should	Inspection
NFR-OE-007	Usability	A first-time Student shall be able to locate and start an assigned exam within 3 clicks from the dashboard and without training.	Must	Usability test (5 users)
NFR-OE-008	Usability	The user interface shall conform to WCAG 2.1 Level AA for colour contrast and keyboard navigation on the exam-taking pages.	Should	Automated audit + inspection
NFR-OE-009	Compatibility	The system shall work correctly on the latest two major versions of Google Chrome, Mozilla Firefox and Microsoft Edge, and shall render usably at screen widths from 360 px upwards.	Must	Cross-browser test

ID	Category	Requirement	Priority	Verification
NFR-OE-010	Timing accuracy	The exam timer and all time-window checks shall use the server clock, with a maximum deviation of 2 seconds between displayed and server time.	Must	Test
NFR-OE-011	Scalability	The system shall store at least 10,000 questions, 1,000 users and 200 exams without degradation of the response-time targets in NFR-OE-001.	Should	Load test
NFR-OE-012	Maintainability	The source code shall be modular, with automated unit tests achieving at least 70% line coverage of business logic.	Should	Coverage report
NFR-OE-013	Portability	The system shall be deployable on a standard Linux server or container using documented steps that a new team member can complete in 60 minutes or less.	Could	Deployment walkthrough
NFR-OE-014	Data retention	The system shall retain exam attempts and results for at least 1 year and shall allow an Administrator to delete data of a deactivated user on request.	Should	Inspection

3.5 Security Requirements

3.5.1 Security Objectives

The following security objectives apply to the Online Exam Portal. Each objective is realised by one or more security requirements in 3.5.2.

ID	Objective	Description
SO-OE-001	Confidentiality	Only authorised users can read exam content, correct answers, personal data and results; correct answers stay hidden from Students until results are released.
SO-OE-002	Integrity	Questions, submitted answers, scores and audit logs cannot be altered by anyone other than the authorised role, and any change is detectable.
SO-OE-003	Authentication & Authorisation	Every user is reliably identified, and every action is restricted to what the user's role permits.
SO-OE-004	Availability	Exams remain accessible to authorised Students during their time window, even under abusive or high traffic.
SO-OE-005	Accountability & Non-repudiation	Security-relevant actions are recorded with user, time and outcome so that incidents can be investigated.

3.5.2 Security Requirements

ID	Objective	Requirement	Priority	Verification
SEC-OE-001	SO-OE-003	The system shall store passwords only as salted hashes using bcrypt or Argon2; plain-text passwords shall never be stored or logged.	Must	Code review + DB inspection
SEC-OE-002	SO-OE-003	The system shall require passwords of at least 8 characters containing upper-case, lower-case and a digit.	Must	Test

ID	Objective	Requirement	Priority	Verification
SEC-OE-003	SO-OE-003	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Must	Test
SEC-OE-004	SO-OE-003	The system shall enforce role-based access control on the server for every request; a request outside the user's role shall return HTTP 403.	Must	Penetration / API test
SEC-OE-005	SO-OE-001	The system shall transmit all data over HTTPS using TLS 1.2 or higher and shall reject plain HTTP.	Must	Scan
SEC-OE-006	SO-OE-003	The system shall expire a session after 30 minutes of inactivity, and session tokens shall be marked HttpOnly and Secure.	Must	Test
SEC-OE-007	SO-OE-001	The system shall never send the correct answers of an exam to the client before the attempt has been submitted and results released.	Must	API response inspection
SEC-OE-008	SO-OE-002	The system shall validate and sanitise all user input and protect against NoSQL/query injection, cross-site scripting and CSRF as listed in the OWASP Top 10.	Must	Security scan + test
SEC-OE-009	SO-OE-002	The system shall accept an answer submission only from the authenticated owner of the attempt and only before the attempt end time (plus a 5-second grace period).	Must	API test
SEC-OE-010	SO-OE-004	The system shall limit login and password-reset requests to 10 per minute per IP address and return HTTP 429 beyond that limit.	Should	Test
SEC-OE-011	SO-OE-005	The system shall log login successes and failures, role changes, exam publication and result access with user ID, timestamp and IP address, and retain these logs for at least 12 months.	Must	Inspection
SEC-OE-012	SO-OE-001	The system shall encrypt personal data (email, name) and backups at rest.	Should	Inspection

3.5.3 Assets and Threats (summary)

Asset	Main threat	Addressed by
Exam questions and correct answers	Leakage before or during exam; Student reads answers from API responses	SEC-OE-004, SEC-OE-007
Student answers and scores	Tampering, late or forged submission, impersonation	SEC-OE-004, SEC-OE-006, SEC-OE-009
User credentials	Brute force, credential theft, weak passwords	SEC-OE-001, SEC-OE-002, SEC-OE-003, SEC-OE-010
Personal data and backups	Interception, unauthorised disclosure	SEC-OE-005, SEC-OE-012
Service availability during exams	Request flooding, abusive traffic	SEC-OE-010, NFR-OE-001, NFR-OE-004
Audit trail	Missing or altered evidence of misuse	SEC-OE-011

3.6 Data Requirements

The system shall persist the following main entities: User (id, name, email, password hash, role, status), Question (id, text, options, correct option, marks, owner), Exam (id, title, duration, start, end, pass mark, settings), ExamQuestion, Assignment (exam, student), Attempt (id, exam, student, start, end, status, score), Answer (attempt, question, selected option), FocusEvent (attempt, timestamp), Notification and AuditLog. In MongoDB these are stored as collections; the detailed schema is defined in the SAD.

4. Initial Requirements Traceability Matrix (RTM)

The RTM links every requirement to its source use case, to the test cases of the Software Test Plan (TC-OE-001 to TC-OE-010) and to the responsible architecture components of the Software Architecture & Design Specification. Design and API references are in the spreadsheet version of this matrix (RTM_Online_Exam_Portal.xlsx).

Req ID	Type	Use case	Priority	Test case ID(s)	Architecture component(s)
FR-OE-001	Functional	UC-01	Must	TC-OE-001	Authentication & Authorization Service
FR-OE-002	Functional	UC-01	Must	TC-OE-001	Authentication & Authorization Service + Notification Service
FR-OE-003	Functional	UC-02	Must	TC-OE-001	Authentication & Authorization Service
FR-OE-004	Functional	UC-02	Must	TC-OE-001	Authentication & Authorization Service
FR-OE-005	Functional	UC-03	Must	TC-OE-002	Authentication & Authorization Service + Notification Service
FR-OE-006	Functional	UC-04	Must	TC-OE-003	Authentication & Authorization Service
FR-OE-007	Functional	UC-05	Must	TC-OE-004	Question Bank Service
FR-OE-008	Functional	UC-06	Must	TC-OE-004	Exam Management Service
FR-OE-009	Functional	UC-06	Should	TC-OE-004	Exam Management Service
FR-OE-010	Functional	UC-06	Could	TC-OE-004	Exam Management Service + Evaluation Engine
FR-OE-011	Functional	UC-07	Must	TC-OE-005	Exam Management Service + Notification Service
FR-OE-012	Functional	UC-06	Must	TC-OE-004	Exam Management Service
FR-OE-013	Functional	UC-08	Must	TC-OE-005	Exam Management Service
FR-OE-014	Functional	UC-09	Must	TC-OE-006	Exam Management Service
FR-OE-015	Functional	UC-09	Must	TC-OE-006	Exam Management Service + Presentation Layer
FR-OE-016	Functional	UC-09	Must	TC-OE-006	Exam Management Service
FR-OE-017	Functional	UC-09	Must	TC-OE-007	Exam Management Service
FR-OE-018	Functional	UC-09	Should	TC-OE-007	Exam Management Service
FR-OE-019	Functional	UC-10	Must	TC-OE-006, TC-OE-008	Exam Management Service
FR-OE-020	Functional	UC-10	Must	TC-OE-006, TC-OE-008	Exam Management Service

Req ID	Type	Use case	Priority	Test case ID(s)	Architecture component(s)
FR-OE-021	Functional	UC-09	Should	TC-OE-006	Exam Management Service + Audit & Logging Service
FR-OE-022	Functional	UC-10	Must	TC-OE-008	Evaluation Engine
FR-OE-023	Functional	UC-11	Must	TC-OE-009	Result & Analytics Service
FR-OE-024	Functional	UC-12	Must	TC-OE-009	Result & Analytics Service
FR-OE-025	Functional	UC-12	Should	TC-OE-009	Result & Analytics Service
FR-OE-026	Functional	UC-07	Should	TC-OE-005	Notification Service
FR-OE-027	Functional	UC-13	Should	TC-OE-003, TC-OE-009	Audit & Logging Service
NFR-OE-001	Non-functional	—	Must	TC-OE-010	All layers (esp. API + MongoDB)
NFR-OE-002	Non-functional	—	Must	TC-OE-007, TC-OE-010	Exam Management Service + Data Access
NFR-OE-003	Non-functional	—	Must	TC-OE-008, TC-OE-010	Evaluation Engine
NFR-OE-004	Non-functional	—	Must	TC-OE-010	Deployment + Monitoring
NFR-OE-005	Non-functional	—	Must	TC-OE-007	Exam Management Service
NFR-OE-006	Non-functional	—	Should	TC-OE-009	MongoDB + Deployment (daily backup job)
NFR-OE-007	Non-functional	—	Must	TC-OE-005	Presentation Layer (React)
NFR-OE-008	Non-functional	—	Should	TC-OE-006	Presentation Layer (React)
NFR-OE-009	Non-functional	—	Must	TC-OE-006	Presentation Layer (React)
NFR-OE-010	Non-functional	—	Must	TC-OE-006	Exam Management Service
NFR-OE-011	Non-functional	—	Should	TC-OE-010	Data Access Layer + MongoDB (indexes)
NFR-OE-012	Non-functional	—	Should	TC-OE-009, TC-OE-010	All Business Services (modular layered structure)
NFR-OE-013	Non-functional	—	Could	TC-OE-010	Deployment (Docker Compose: React, Node.js, MongoDB)
NFR-OE-014	Non-functional	—	Should	TC-OE-009	Authentication & Authorization Service + Data Access Layer + MongoDB
SEC-OE-001	Security	—	Must	TC-OE-001	Authentication & Authorization Service
SEC-OE-002	Security	—	Must	TC-OE-001	Authentication & Authorization Service
SEC-OE-003	Security	—	Must	TC-OE-002	Authentication & Authorization Service
SEC-OE-004	Security	—	Must	TC-OE-003	Authentication & Authorization Service
SEC-OE-005	Security	—	Must	TC-OE-001	API Layer + Infrastructure

Req ID	Type	Use case	Priority	Test case ID(s)	Architecture component(s)
SEC-OE-006	Security	—	Must	TC-OE-001	Authentication & Authorization Service
SEC-OE-007	Security	—	Must	TC-OE-006	Exam Management + Evaluation Engine
SEC-OE-008	Security	—	Must	TC-OE-004	API Layer + All Services
SEC-OE-009	Security	—	Must	TC-OE-006	Exam Management Service
SEC-OE-010	Security	—	Should	TC-OE-002	API Layer
SEC-OE-011	Security	—	Must	TC-OE-003, TC-OE-005, TC-OE-009	Audit & Logging Service
SEC-OE-012	Security	—	Should	TC-OE-009	Data Access Layer + MongoDB

Coverage summary: 27 functional, 14 non-functional and 12 security requirements (53 in total), derived from 13 use cases and 5 security objectives.

5. Appendices

Appendix A – Requirement ID Convention

Type	Format	Example
Functional requirement	FR-OE-NNN	FR-OE-014
Non-functional requirement	NFR-OE-NNN	NFR-OE-001
Security requirement	SEC-OE-NNN	SEC-OE-004
Security objective	SO-OE-NNN	SO-OE-001
Use case	UC-NN	UC-09
Test case (owned by P2)	TC-OE-NNN	TC-OE-001

IDs are never reused or renumbered after freeze. A removed requirement is marked “Deleted” rather than removed from the document.

Appendix B – Resolved Issues

#	Issue	Resolution
1	Technology stack	MERN stack fixed (section 2.5).
2	Test case IDs in the RTM	Taken from Software Test Plan v1.1 (TC-OE-001 to TC-OE-010); every requirement is covered by at least one test case.
3	Architecture components in the RTM	Taken from SAD section 3.8 and the component diagram; NFR-OE-006 to 009 and 011 to 014 were mapped from the SAD design sections.
4	Result release and question-wise review (FR-OE-023)	Score shown immediately after submission; correct-answer review only after the exam end date-time (section 2.6). No open issues remain.